AIM: The aim of this project is to identify the factors that contribute to the success of a movie, specifically whether the production company, the movie's budget, or other variables play a significant role in determining a movie's success. By performing a data analysis using Python, this project aims to provide valuable insights that can help movie production companies make more informed decisions about which movies to produce, which companies to partner with, and how to allocate resources to maximize the chances of success

In [6]:
```
import pandas as pd

# Load both datasets
movies_df = pd.read_csv('tmdb_5000_movies.csv')
credits_df = pd.read_csv('tmdb_5000_credits.csv')
import os
os.getcwd()
```

Out[6]: 'C:\\Users\\shubhangi singh\\movie_sucess analysis'

In [37]:
```
import pandas as pd

movies_df = pd.read_csv(r'C:\Users\shubhangi singh\movie_sucess analysis\tmdb_5000_movies.csv')
credits_df = pd.read_csv(r'C:\Users\shubhangi singh\movie_sucess analysis\tmdb_5000_credits.csv')
```

In [11]:
```
# Flatten all genres and count the frequency
from collections import Counter

# Flatten the genres into a single list
all_genres = [genre for sublist in movie_data['genres'] for genre in sublist]
genre_counts = Counter(all_genres)

# Convert the counter to a DataFrame
genre_df = pd.DataFrame(genre_counts.items(), columns=['Genre', 'Count'])
genre_df = genre_df.sort_values(by='Count', ascending=False)

# Preview the result
genre_df.head(10)
```

Out[11]:

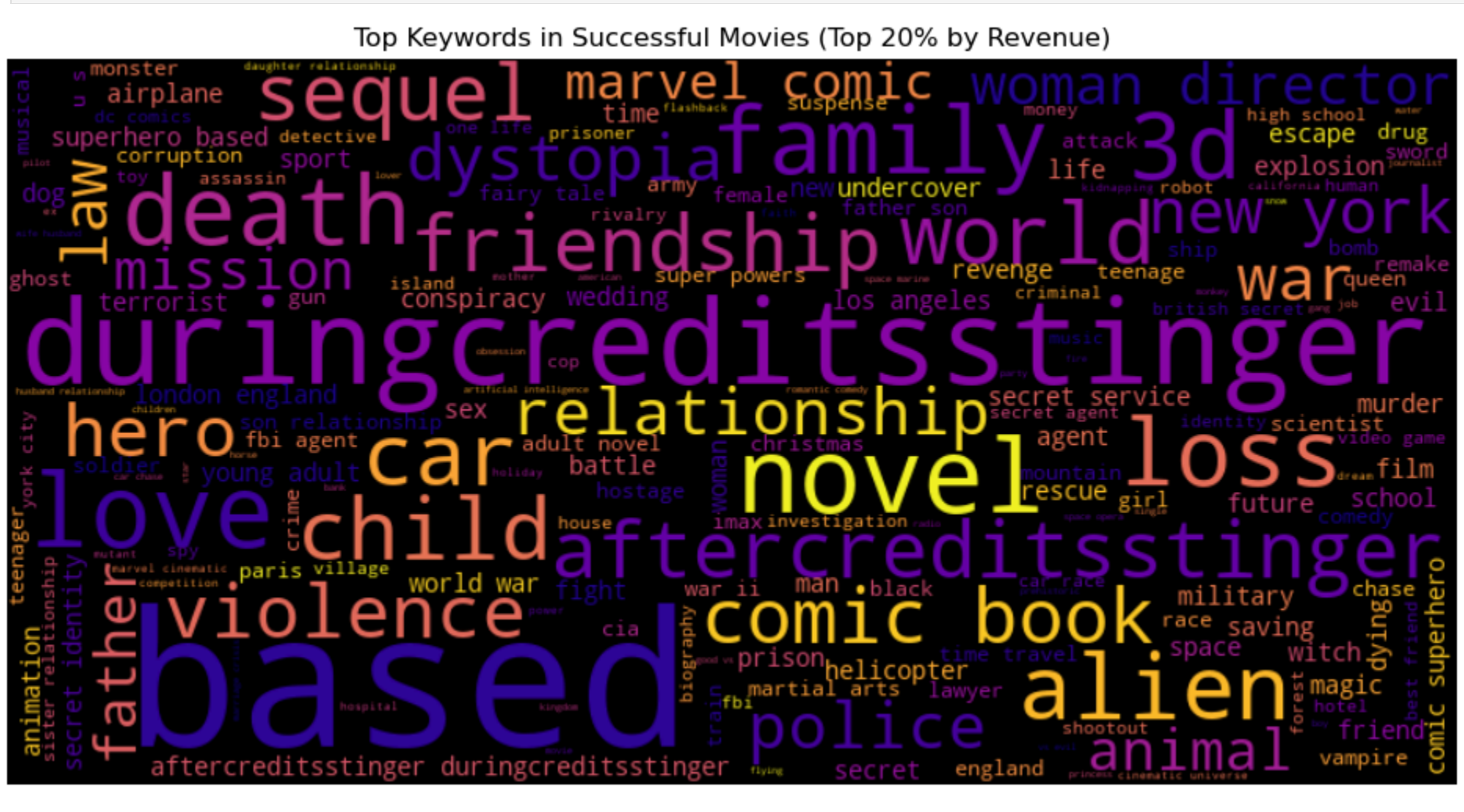
	Genre	Count
2	"	72960
6		44408
5	:	24320
13	e	20060
9	,	19545
11	a	19497
12	m	18626
3	i	18294
10	n	17186
4	d	14672

In [13]:
```
# Calculate ROI (Avoid division by zero)
movie_data['ROI'] = movie_data.apply(lambda x: (x['revenue'] - x['budget']) / x['budget'] * 100 if x['budget'] > 0 else 0, axis=1)

# Preview the result
movie_data[['original_title', 'ROI']].head()
```

Out[13]:

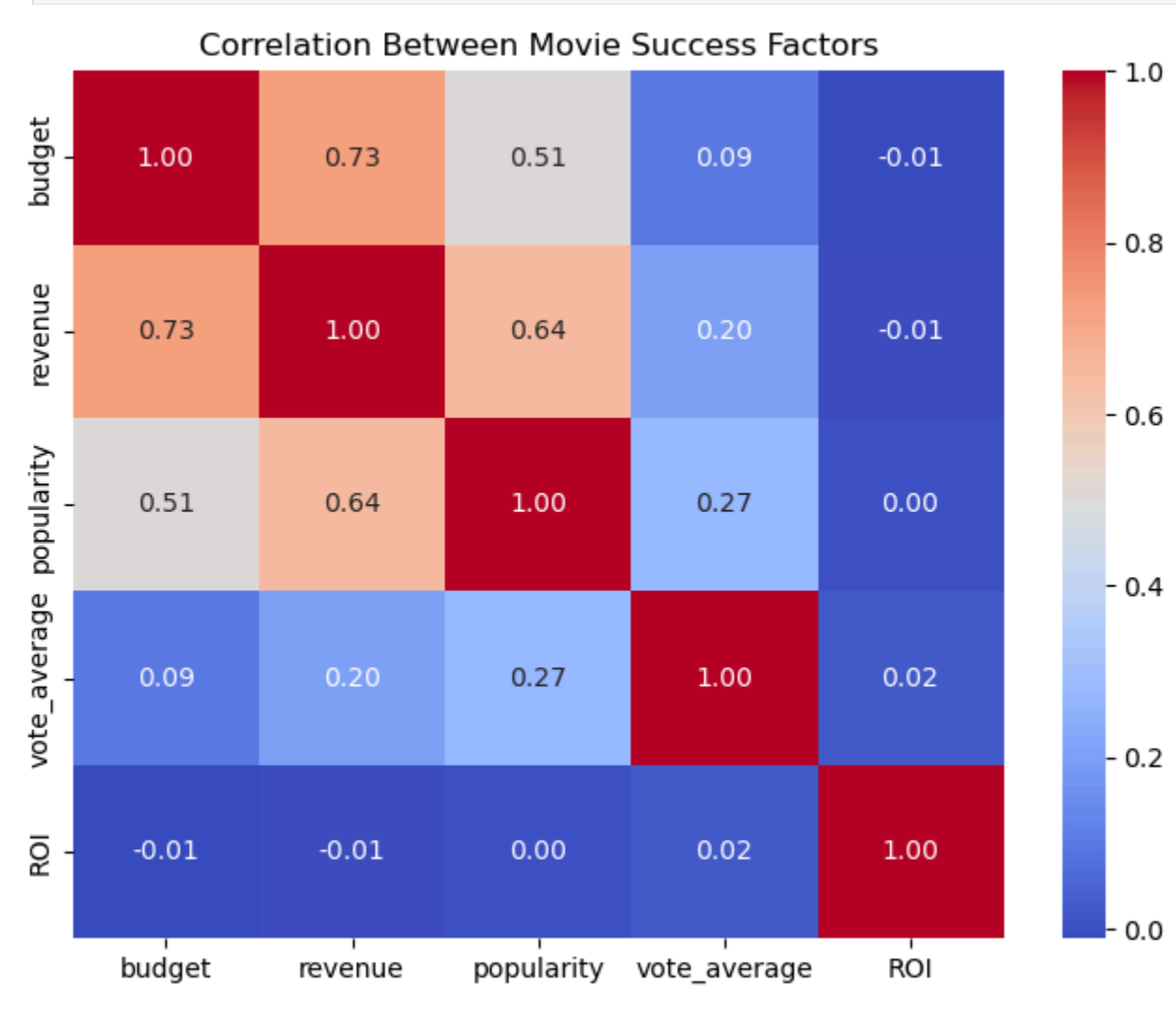
	original_title	ROI
0	Avatar	1076.356577
1	Pirates of the Caribbean: At World's End	220.333333
2	Spectre	259.459024
3	The Dark Knight Rises	333.975640
4	John Carter	9.284269

In [51]:
```
from wordcloud import WordCloud
import matplotlib.pyplot as plt
import ast

# Convert 'keywords' to list using ast
movie_data['keywords_list'] = movie_data['keywords'].dropna().apply(lambda x: [d['name'] for d in ast.literal_eval(x)] if x != '[]' else [])

# Define "successful" movies: Top 20% by revenue
threshold = movie_data['revenue'].quantile(0.80)
successful_movies = movie_data[movie_data['revenue'] >= threshold]

# Combine all keywords of successful movies into one string
successful_keywords = successful_movies['keywords'].dropna().apply(lambda x: ' '.join([d['name'] for d in ast.literal_eval(x)])).str.cat(sep=' ')

# Generate word cloud
wordcloud = WordCloud(width=800, height=400, background_color='black', colormap='plasma').generate(successful_keywords)

# Plot it
plt.figure(figsize=(12, 6))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title('Top Keywords in Successful Movies (Top 20% by Revenue)')
plt.show()
```



In [ ]:
```
Conclusion :This shows the keywords of sucessful movies
```

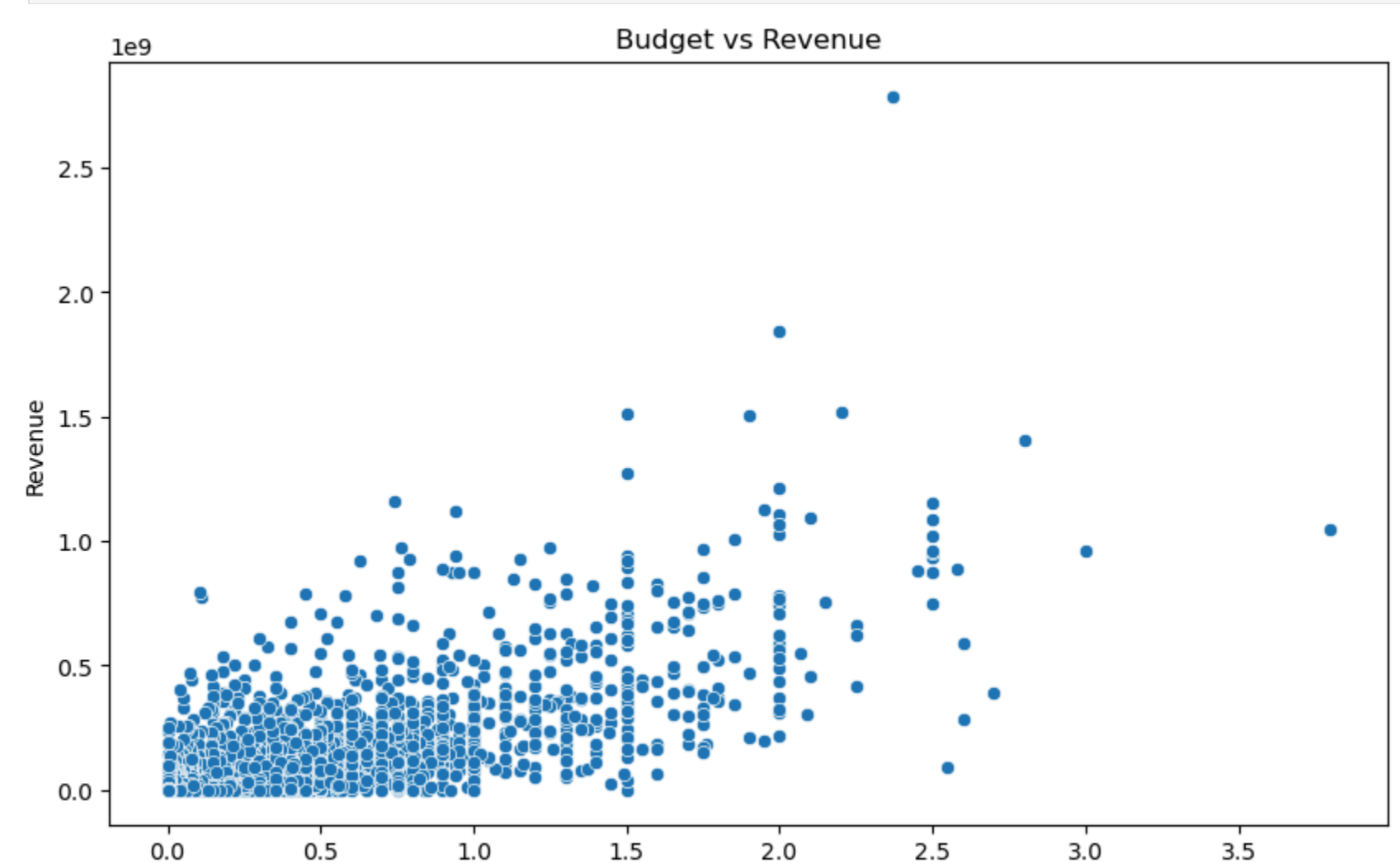
In [16]:
```
# Select numeric columns for correlation
correlation_data = movie_data[['budget', 'revenue', 'popularity', 'vote_average', 'ROI']]

# Compute correlation matrix
correlation_matrix = correlation_data.corr()

# Plot heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f")

plt.title('Correlation Between Movie Success Factors')
plt.show()
```

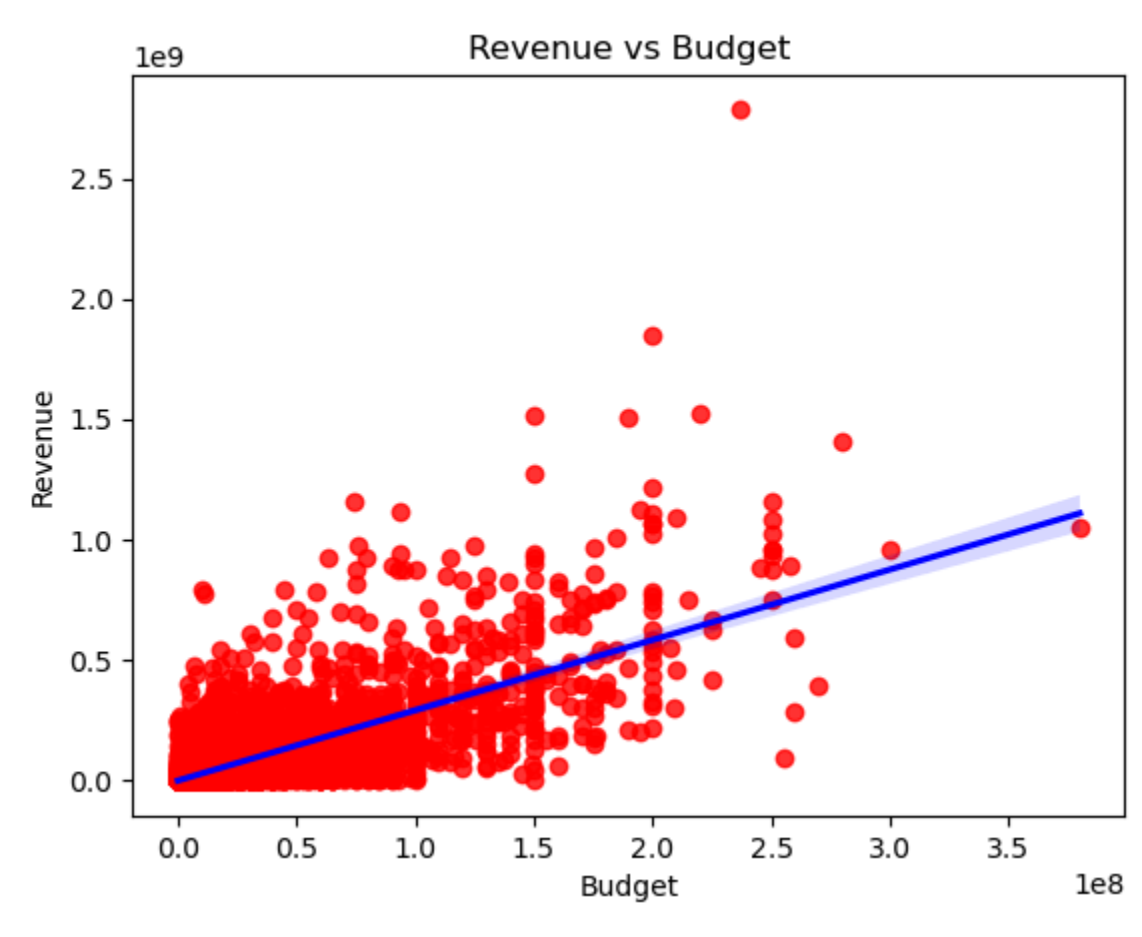


	budget	revenue	popularity	vote_average	ROI
budget	1.00	0.73	0.51	0.09	-0.01
revenue	0.73	1.00	0.64	0.20	-0.01
popularity	0.51	0.64	1.00	0.27	0.00
vote_average	0.09	0.20	0.27	1.00	0.02
ROI	-0.01	-0.01	0.00	0.02	1.00

Popularity and vote count are strongly correlated with higher revenue, making them key indicators of movie success. In contrast, higher ratings alone do not guarantee financial success. You can see more detail in below graph also in the "Correlation Matrix for Movie Features".

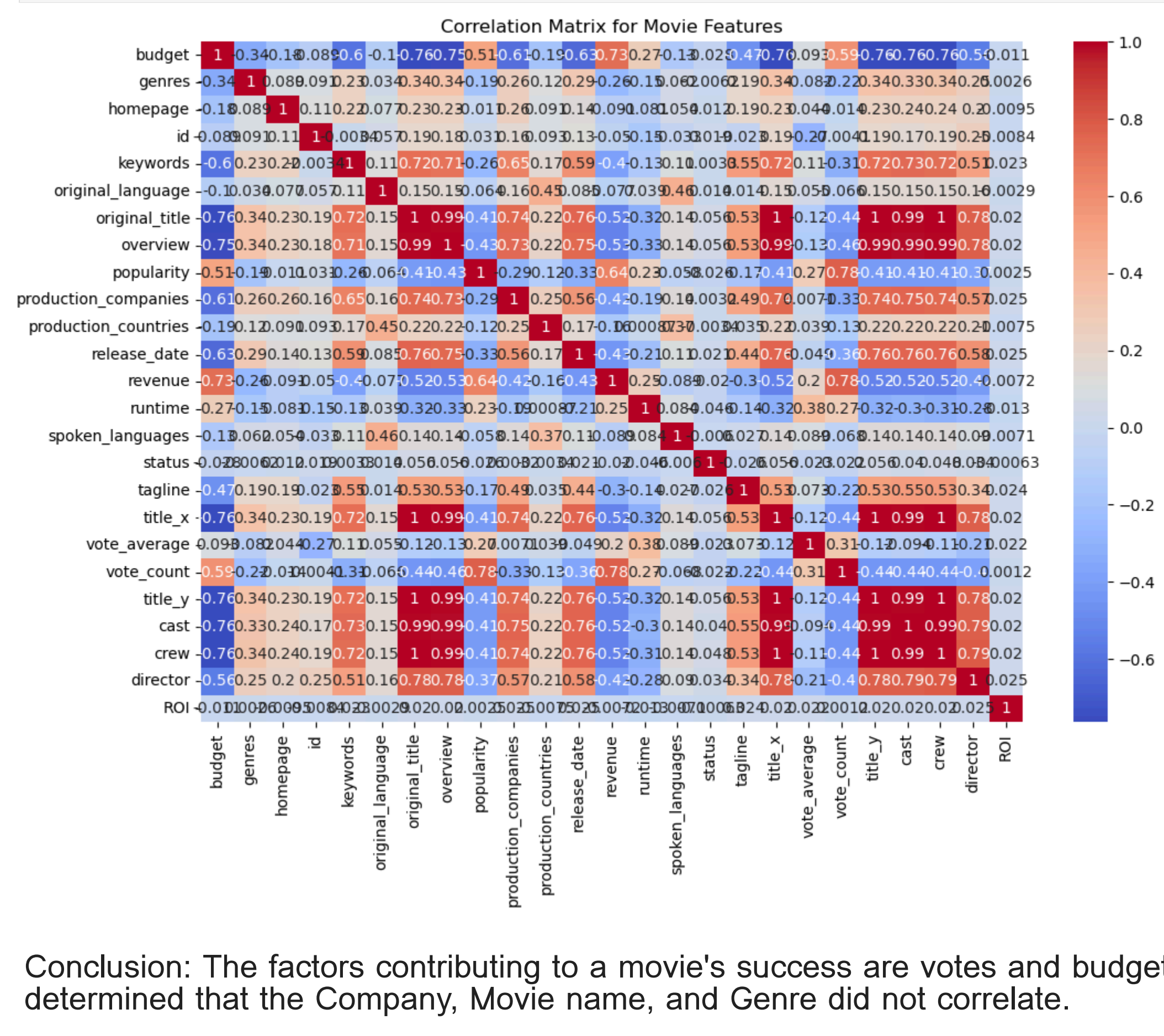
In [21]:
```
plt.figure(figsize=(10,6))
sns.scatterplot(x='budget', y='revenue', data=movie_data)
plt.title('Budget vs Revenue')
plt.xlabel('Budget')
plt.ylabel('Revenue')
plt.show()

# Looking at correlation
sns.regplot(x="budget", y="revenue", data=movie_data, scatter_kws={"color": "red"}, line_kws={"color": "blue"}).set_title('Revenue vs Budget')
plt.xlabel('Budget')
plt.ylabel('Revenue')
plt.show()
```





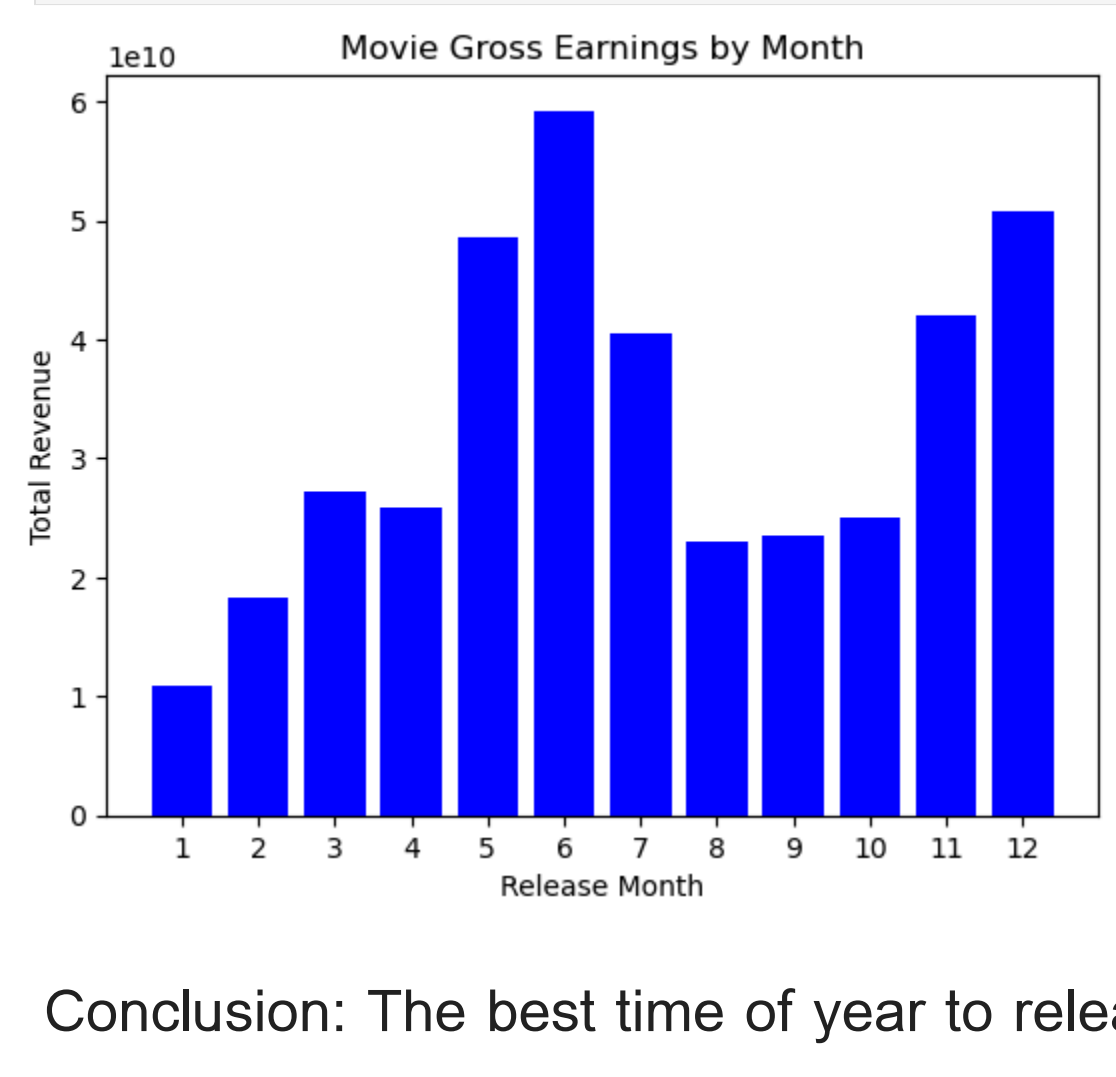
Conclusion: From this graph, we can determine that there is a significant relationship between the budget and gross earnings of a movie.

In [23]:
```
import numpy as np

# Make a copy of your main DataFrame
df_numerized = movie_data.copy()

# Convert object (categorical) columns to numeric codes
for col in df_numerized.columns:
    if df_numerized[col].dtype == 'object':
        df_numerized[col] = pd.factorize(df_numerized[col])[0]
correlation_matrix = df_numerized.corr(method='pearson')

plt.figure(figsize=(12, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title("Correlation Matrix for Movie Features")
plt.show()
```



Conclusion: The factors contributing to a movie's success are votes and budget, as they have the highest correlation. It was also determined that the Company, Movie name, and Genre did not correlate.

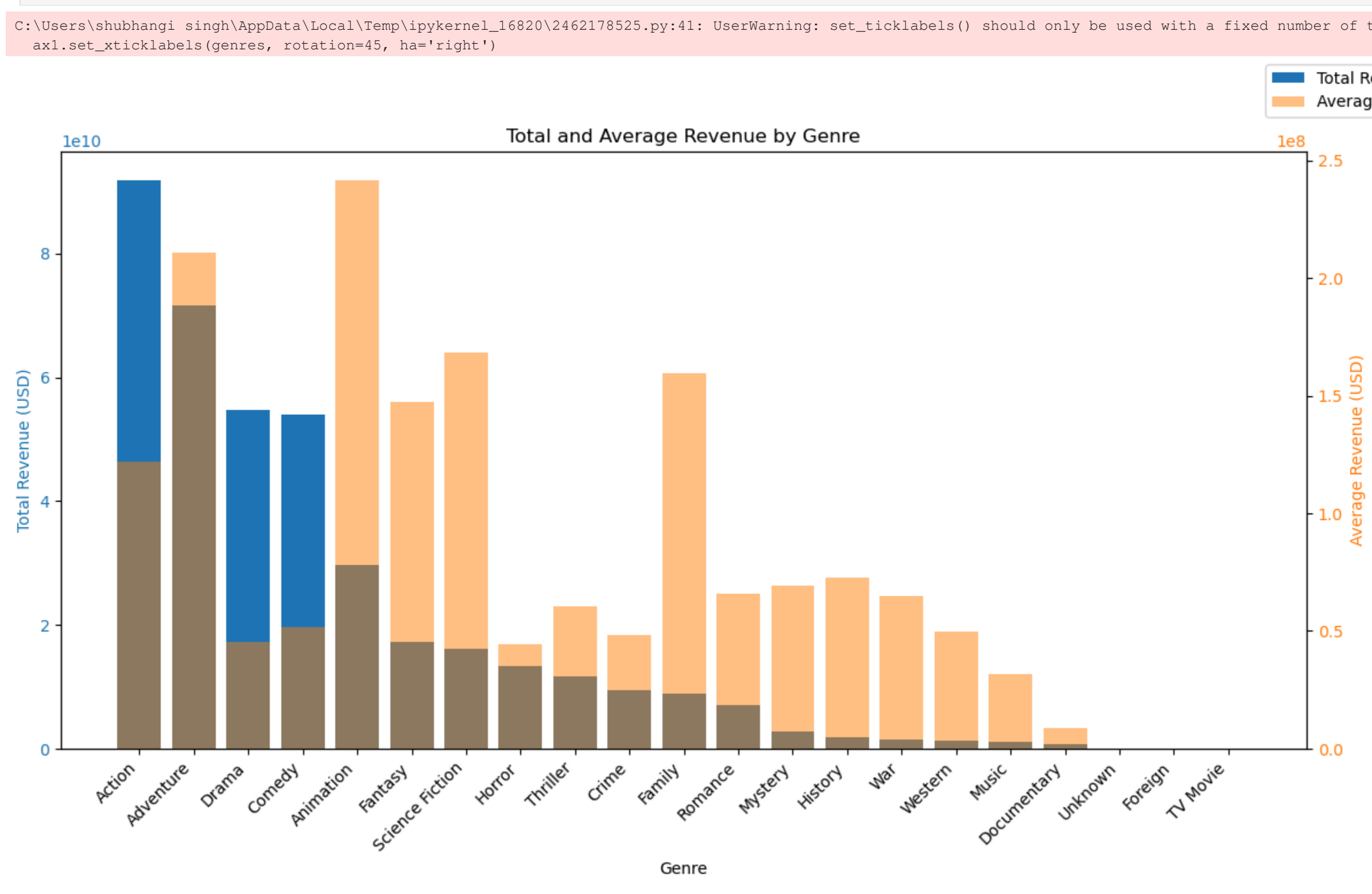
In [26]:
```
# Extract month from release date
movie_data['release_date'] = pd.to_datetime(movie_data['release_date'], errors='coerce')
movie_data['release_month'] = movie_data['release_date'].dt.month

#Plot gross earnings by month
months = range(1, 13)

plt.bar(months, movie_data.groupby(['release_month'])['revenue'].sum(), color='blue')
plt.xticks(months)
plt.xlabel("Release Month")
plt.ylabel("Total Revenue")
plt.title("Movie Gross Earnings by Month")
plt.show()
```



Conclusion: The best time of year to release a movie is june based on the overall gross earnings in this period

In [50]:
```
import pandas as pd
import matplotlib.pyplot as plt
import ast

# Load datasets
movies_df = pd.read_csv(r'C:\Users\shubhangi singh\movie_sucess analysis\tmdb_5000_movies.csv')
credits_df = pd.read_csv(r'C:\Users\shubhangi singh\movie_sucess analysis\tmdb_5000_credits.csv')

# Merge on 'id'
credits_df.rename(columns={'movie_id': 'id'}, inplace=True)
movie_data = movies_df.merge(credits_df, on='id')

# Simplify the 'genres' column (extract first genre)
def extract_genre(genre_str):
    try:
        genre_list = ast.literal_eval(genre_str)
        return genre_list[0]['name'] if genre_list else 'Unknown'
    except:
        return 'Unknown'

movie_data['genre'] = movie_data['genres'].apply(extract_genre)

# Group by genre and calculate sum and mean of revenue
grouped_df = movie_data.groupby('genre')['revenue'].agg(['sum', 'mean']).sort_values(by='sum', ascending=False)

# Prepare data for plotting
genres = grouped_df.index.tolist()
sum_revenue = grouped_df['sum'].tolist()
mean_revenue = grouped_df['mean'].tolist()

# Plotting
fig, ax1 = plt.subplots(figsize=(14, 8))  # Larger figure

# Bar plot for total revenue
color = 'tab:blue'
ax1.bar(genres, sum_revenue, color=color, label='Total Revenue')
ax1.set_xlabel('Genre')
ax1.set_ylabel('Total Revenue (USD)', color=color)
ax1.tick_params(axis='y', labelcolor=color)
ax1.set_title('Total and Average Revenue by Genre')
ax1.set_xticklabels(genres, rotation=45, ha='right')

# Second axis for average revenue
ax2 = ax1.twinx()
color = 'tab:orange'
ax2.bar(genres, mean_revenue, color=color, alpha=0.5, label='Average Revenue')
ax2.set_ylabel('Average Revenue (USD)', color=color)
ax2.tick_params(axis='y', labelcolor=color)

# Add legends
fig.legend(loc='upper right')

# Manual layout adjustment
plt.subplots_adjust(bottom=0.25, top=0.9)

plt.show()
```

```
C:\Users\shubhangi singh\AppData\Local\Temp\ipykernel_16820\2462178525.py:41: UserWarning: set_ticklabels() should only be used with a fixed number of ticks, i.e. after set_ticks() or using a FixedLocator.
  ax1.set_xticklabels(genres, rotation=45, ha='right')
```



Conclusion: The Action and animation genre had the highest level of success across all movies, whereas the Family genre had the highest average success rate.

Conclusion: My analysis revealed that the most significant factors contributing to a movie's success were the production, the movie's budget, and the votes. Additionally, certain genres, such as action and family, tended to be more successful than others.

Overall, my Python data analysis project provided valuable insights into the factors contributing to a movie's success.